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| **Fiscal Year:** | FY 2021 | **Task Last Updated:** | FY 12/08/2020 |
| **PI Name:** | Durian, Douglas J. Ph.D. | | |
| **Project Title:** | Foam Optics and Mechanics | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | COMPLEX FLUIDS/SOFT MATTER--Complex Fluids | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 19104-6396 | **Congressional District:** | 2 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 1995 Microgravity Fluid Physics NRA-94-OLMSA-05: Research and Flight Experiment Opportunities |
| **Start Date:** | 02/20/2019 | **End Date:** | 11/30/2020 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA GRC |
| **Contact Monitor:** | Chao, David | **Contact Phone:** | 216-433-8320 |
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| **Flight Program:** | ISS | | |
| **Flight Assignment:** | NOTE: End date changed to 11/30/2020 per NSSC information (Ed., 9/9/20) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | 80NSSC19K0599 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | Continuation of "FOAM--Foam Optics and Mechanics (The Melting of Aqueous Foams)," grant NNX14AM99G, and previously "The Melting of Aqueous Foams," grant numbers NNX07AP20G ; NNX07AK60G ; NNC04GB61G.<br>The unusual elastic character of foams will be quantified macroscopically by measurement of the shear stress as a function of static shear strain, shear strain rate, and time following a step strain; such data will be analyzed in terms of a yield stress, a static shear modulus, and dynamical time scales. Microscopic information about bubble packing and rearrangement dynamics, from which these macroscopic non-Newtonian properties presumably arise, will be obtained non-invasively by novel multiple-light scattering diagnostics such as diffusing-wave spectroscopy (DWS) and speckle-visibility spectroscopy (SVS). Bubble size and homogeneity will be studied by diffusely transmitted light (DTS). Liquid fraction will be measured by electrical conductivity. Quantitative trends with materials parameters, such as average bubble size, and liquid content, will be sought in order to elucidate the fundamental connection between the microscopic structure and dynamics and the macroscopic rheology.<br><br>There are three all-encompassing objectives motivating the FOAM project. The first concerns multiphase flows and represents the “mechanics” aspect alluded to in the F.O.A.M. acronym: To understand how foams and bubbly liquids deform and flow in response to applied forces. The second objective concerns multiphase separations: To understand how foams and bubbly liquids evolve with time via a combination of liquid flow between bubbles, of gas diffusion across the film separating neighboring bubbles, and of direct coalescence via film rupture. For both flows and separations, some level of understanding has been established for very dry foams, and also for liquids with relatively few bubbles; however, we are more ignorant of the crucial intermediate regime of very wet foams and very bubbly liquids. This regime defies Earth-bound experiments. The third, overall, objective is to transfer this new understanding of multiphase flows and separations into Advanced Life Support (ALS) technology for NASA's strategic exploration mission.<br><br>The utility and fascination of foams are derived largely from the surprising fact that they have a solid-like elastic character in spite of being mostly gas with a few percent volume fraction of liquid, but can nevertheless flow under shear. The physical origin of such unusual rheology in terms of microscopic structure and dynamics is poorly understood and remains a subject of basic scientific interest to physicists, chemists, and chemical engineers. The proposed research promises important new insight into these issues, and could also have significant consequences for our understanding of flow in other dense randomly-packed systems such as emulsions, colloidal suspensions, slurries, bubbly liquids, and granular materials. Furthermore, all foam applications are empirically based and the proposed research may generate valuable fundamental guidance for the development of materials with more desirable rheology and better stability.<br><br>The value of the FOAM project in terms of NASA’s exploration mission lies with Advanced Life Support. Water recovery and recycling are absolutely central to ALS, and require fine control of multiphase flows and separations. The very heart of the FOAM project is to bring new technology and understanding to these problems. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Foams are important for a clean environment. Firstly, their presence in natural waters signals the presence of surface-active impurities and a need for clean-up. Secondly, they are useful themselves in the cleanup process since many impurities can be removed by injecting gas and skimming off the resulting foam. Relatedly, study of foams brings specific insight into multiphase flow and separation phenomena of direct importance for water recovery and recycling operations. |
| **Task Progress:** | During this cycle my FOAM group consisted of Anthony Chieco, who began as a PhD student and finished as a postdoc. One focus of our work was on ground-based studies of the structure and time evolution of coarsening foams, in support of the in-progress and planned flight experiments with experiment container SOFT MATTER DYNAMICS (formerly FOAM-C). This program is being financed and directed by the European Space Agency. The instrument itself is aboard the International Space Station (ISS) and the first measurement campaign successfully concluded in November 2020; the second campaign is scheduled for Summer-through-Fall 2021. The first round of sample cells was devoted to the coarsening of foams, the fluid/glassy dynamics of granular matter, and fundamentals of light scattering. The FOAM focus is on structure, stability, and temporal evolution, for sequences of progressively wetter foams. These cannot be studied on Earth due to gravitation drainage, which increases with wetness and effectively prevents study of behaving when the bubbles become nearly spherical and unjammed.<br>At Penn, we have made progress in several areas during the current funding period. This includes theoretical work on how gas is transported across Plateau borders in wet foams. In the past, borders have been assumed to totally block the gas; earlier we showed that much more is transported than naïve scaling expectations, and this forms the basis for new/unpublished predictions now being tested. Work also includes the application of our new way to search for hidden order in amorphous systems, using a new concept for how diagnose hyperuniformity in terms of the length scale over which fluctuations are important, to a coarsening foam in self-similar state as well as a host of other cellular systems for comparison. |
| **Bibliography Type:** | Description: (Last Updated: 06/12/2025) |
| **Abstracts for Journals and Proceedings** | Chieco AT, Durian DJ. "Quantifying the Structure of Space-Filling Disordered Cellular Patterns with Hyperuniformity Disorder Length Spectroscopy." Presented at the APS March Meeting 2019, Boston, MA, March 4-8, 2019.<br>Bulletin of the American Physical Society. 2019 Mar;64(2):Abstract. K58.00012.<br>https://meetings.aps.org/Meeting/MAR19/Session/K58.12 , Mar-2019 |
| **Abstracts for Journals and Proceedings** | Durian DT, Chieco AT. "Experimental Test of the Border-Crossing Model of Diffusive Coarsening in Wet Foams." Invited presentation at APS March Meeting 2020, Denver, CO, March 2–6, 2020 (cancelled).<br>Bulletin of the American Physical Society. 2020 Mar;65(1):Abstract:S33.00005.<br>http://meetings.aps.org/Meeting/MAR20/Session/S33.5 , Mar-2020 |

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| **Abstracts for Journals and Proceedings** | Chieco AT, Durian DJ. "How Fluid Accumulation Affects the Dynamics of Bubble Growth and Rearrangements." Presented at the 73rd Annual Meeting of the APS Division of Fluid Dynamics, Virtual, November 22–24, 2020. (online)<br>Bulletin of the American Physical Society. 2020 Nov;65(13):Abstract:R02.00006.<br>https://meetings.aps.org/Meeting/DFD20/Session/R02.6 , Nov-2020 |
| **Articles in Other Journals or Periodicals** | Chieco AT, Durian DJ. "Experimentally testing a generalized coarsening model for individual bubbles in quasi-two-dimensional wet foams." Phys Rev E. Submitted October 13, 2020. In server: https://arxiv.org/abs/2010.06664 , Oct-2020 |
| **Dissertations and Theses** | Chieco AT. "Quantifying Structure in Amorphous Materials at All Length Scales: From Uncovering Long Range Hidden Order With Hyperuniformity Disorder Length Spectroscopy to Using Local Bubble Shape To Predict Foam Aging Dynamics." PhD Dissertation, University of Pennsylvania, May 2019. , May-2020 |